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Question 1

Correct

Mark 0.25 out of
0.25

gRPC suits ride-sharing because:

- ☐ a. No auth
- ☐ b. REST slower
- ☒ c. Bi-directional streaming with low latency ✓
- ☐ d. Shared schema



The correct answer is: Bi-directional streaming with low latency

Question 2

Correct

Mark 0.25 out of
0.25

Kafka is primarily used for:

- ☒ a. Managing high-velocity real-time data streams ✓
- ☐ b. Authentication
- ☐ c. File storage
- ☐ d. API Versioning

The correct answer is: Managing high-velocity real-time data streams

Question 3

Correct

Mark 0.25 out of
0.25

gRPC differs from REST by:

- ☐ a. Using JSON/HTTP1.1
- ☐ b. Centralized ESB
- ☒ c. Using Protobuf over HTTP/2 for streaming ✓
- ☐ d. Stateless cache only

The correct answer is: Using Protobuf over HTTP/2 for streaming

Question 4

Correct

Mark 0.25 out of
0.25

Which is a disadvantage of monolithic applications?

- ☒ a. Slow development and long deployment cycles ✓
- ☐ b. Loose coupling
- ☐ c. Simple networking
- ☐ d. Easy testing



The correct answer is: Slow development and long deployment cycles

Question 5

Correct

Mark 0.25 out of
0.25

A drawback of serverless computing which can affect performance are:

- ☐ a. Pay-as-you-go billing
- ☐ b. Unlimited scaling
- ☒ c. Cold start latency & state management issues ✓
- ☐ d. Lower operational costs

The correct answer is: Cold start latency & state management issues

Question 6

Correct

Mark 0.25 out of
0.25

In Netflix's evolution, the critical need was:

- ☐ a. Lower cost DB
- ☒ b. Elasticity and modularity ✓
- ☐ c. Avoid caching
- ☐ d. Minimize API gateways

The correct answer is: Elasticity and modularity

Question 7

Correct

Mark 0.25 out of
0.25

NoSQL databases provide scalability mainly through:

- ☐ a. Disk mirroring
- ☐ b. Vertical scaling
- ☒ c. Horizontal scaling with sharding ✓
- ☐ d. Disk mirroring



The correct answer is: Horizontal scaling with sharding

Question 8

Correct

Mark 0.25 out of
0.25

In FTGO case study, the main problem with the old monolith was:

- ☐ a. Simple DB schema
- ☐ b. Use of external APIs
- ☒ c. Complexity, scaling difficulty, and slow updates ✓
- ☐ d. Too many microservices

The correct answer is: Complexity, scaling difficulty, and slow updates

Question 9

Correct

Mark 0.25 out of
0.25

A key characteristic of microservices is:

- ☒ a. Independent deployment ✓
- ☐ b. Shared schema for all services
- ☐ c. Single runtime environment
- ☐ d. Centralized governance

The correct answer is: Independent deployment

Question 10

Correct

Mark 0.25 out of
0.25

Why is CQRS preferred in e-commerce?

- ☐ a. Avoids replication
- ☐ b. Guarantees strong consistency
- ☒ c. Optimizes read-heavy and write-heavy ops separately ✓
- ☐ d. Reduces need for caching



The correct answer is: Optimizes read-heavy and write-heavy ops separately

Question 11

Correct

Mark 0.25 out of
0.25

Eventual consistency is typically used in:

- ☐ a. Banking systems
- ☒ b. Social media platforms ✓
- ☐ c. File compression tools
- ☐ d. Compiler design

The correct answer is: Social media platforms

Question 12

Correct

Mark 0.25 out of 0.25

The Strangler Pattern is used to:

- ☐ a. Share DB schemas
- ☐ b. Optimize ESB performance
- ☐ c. Scale out databases
- ☒ d. Incrementally replace monoliths ✓

The correct answer is: Incrementally replace monoliths

Question 13

Correct

Mark 0.25 out of 0.25

Horizontal partitioning is also called:

- ☐ a. Layering
- ☒ b. Sharding ✓
- ☐ c. Vertical partitioning
- ☐ d. Functional partitioning



The correct answer is: Sharding

Question 14

Correct

Mark 0.25 out of 0.25

Resilience in microservices is achieved through:

- ☐ a. Ignoring failures
- ☐ b. Shared states
- ☐ c. Tight coupling
- ☒ d. Fault tolerance & monitoring ✓

The correct answer is: Fault tolerance & monitoring

Question 15

Correct

Mark 0.25 out of
0.25

A monolithic application is best described as:

- ☐ a. A collection of microservices
- ☐ b. A modular system with SOA
- ☐ c. A polyglot persistence model
- ☒ d. An application composed all in one piece, tightly coupled ✓

The correct answer is: An application composed all in one piece, tightly coupled

Question 16

Correct

Mark 0.25 out of
0.25

In edge computing, data is processed:

- ☒ a. Near the source of data generation ✓
- ☐ b. After transmission to central servers
- ☐ c. Only in the cloud
- ☐ d. In a single global database



The correct answer is: Near the source of data generation

Question 17

Correct

Mark 0.25 out of
0.25

Replication improves scalability by ensuring:

- ☐ a. Data encryption
- ☒ b. High availability and load sharing ✓
- ☐ c. Monolithic processing
- ☐ d. Single points of failure

The correct answer is: High availability and load sharing

Question 18

Correct

Mark 0.25 out of
0.25

Which of the following is a vertical scalability approach?

- ☐ a. Replicating across data centers
- ☒ b. Adding more CPUs or RAM to a single server ✓
- ☐ c. Sharding
- ☐ d. Adding more servers

The correct answer is: Adding more CPUs or RAM to a single server

Question 19

Correct

Mark 0.25 out of
0.25

Microservices are modeled around:

- ☐ a. Middleware integration
- ☐ b. Programming languages
- ☒ c. Business domains/capabilities ✓
- ☐ d. Shared databases



The correct answer is: Business domains/capabilities

Question 20

Correct

Mark 0.25 out of
0.25

SOA typically differs from microservices by:

- ☐ a. Fine-grained service boundaries
- ☒ b. Relying on ESB with SOAP/XML ✓
- ☐ c. Using lightweight REST APIs
- ☐ d. Decentralized governance

The correct answer is: Relying on ESB with SOAP/XML

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Answered: 100% (10/10) Correct: 10/10 (100%)
Time taken: 00:00:00 (00:00:00)

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